

Software Requirements Specification

Reconciliation & Reporting Tool

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the detailed functional, validation, business, workflow, and non-functional requirements for the Reconciliation & Reporting Tool. The document is intended to guide business stakeholders, project managers, designers, developers, QA engineers, and deployment teams so the solution can be implemented with minimal ambiguity.

1.2 Background

Client operates with seven OpCos and more than 35 partners, manages multiple revenue streams, issues invoices to OpCos, receives invoices from partners, collects payments from OpCos, and settles partner payments. The required system is a centralized web application to manage these reconciliation and reporting activities in a structured, automated, and auditable way.

1.3 Scope

The application shall support:

- secure login and user access
- report upload and viewing for OpCos and partners
- invoice upload, generation, tracking, and status updates
- reconciliation of OpCo and partner reports
- mismatch handling and confirmation workflows
- email notifications and reminders
- dashboard, search, sorting, filtering, export, and adhoc reporting
- administration, user management, and settings
- future ERP integration with Oracle EBS / Oracle Fusion at a later phase

These expectations are directly aligned with the scope and technical features in the source document.

1.4 Product Type

The system shall be a web-based, responsive, English-language application.

2. Definitions and Business Terms

2.1 Definitions

Client

The internal operating entity using the platform for reconciliation and financial coordination.

OpCo

An operating company that submits reports and receives invoices from Client.

Partner

An external partner that submits reports and invoices and receives payments from Client.

Reporting Period

A month-year cycle used for submission, reconciliation, and settlement.

Service

A revenue stream or service category. Initial examples include DoB, Digital Advertising, and API Marketplace.

Reconciliation Key

The business identifier used to compare records. At minimum this shall include Reporting Month, OpCo, Partner where applicable, and Service.

RS Report

Final confirmation report shared with partners after reconciliation, as referenced in the scope.

Collection Received

Confirmation that payment from the relevant OpCo has been received by Client.

3. User Roles

3.1 Admin

Responsible for platform administration, user management, notification configuration, reminder settings, and platform-level controls.

3.2 Client User

Responsible for reconciliation operations, report and invoice review, status updates, mismatch handling, confirmation, search, export, dashboards, reminders, and communication with OpCos and partners.

3.3 OpCo User

Responsible for uploading monthly reports, viewing own submitted reports, viewing invoices issued to their OpCo, and monitoring invoice/payment status relevant to their entity.

3.4 Partner User

Responsible for uploading monthly reports, uploading partner invoices when allowed, viewing submitted reports and invoices, and monitoring invoice/payment status relevant to their entity.

4. Overall Functional Modules

The solution shall contain the following modules, aligned with the provided scope:

1. Authentication and Access Control
2. Dashboard
3. Report Management
4. Invoice Management
5. Reconciliation Management
6. Payment and Collection Tracking
7. Search, Filter, Sort, and Export
8. Notification and Reminder Management
9. Administration
10. Audit and Activity Logging

- 11. Master Data and Configuration
- 12. Future ERP Integration Layer

The scope explicitly lists common functionalities, OpCo/Partner module, Client-Zain module, Administration module, dashboards, reminders, export, and future Oracle connectivity.

5. Assumptions

- 1. Initial application language shall be English only.
 - 2. Each report and invoice belongs to a defined reporting month and year.
 - 3. Client shall maintain master data for OpCos, partners, services, currencies, and notification templates.
 - 4. Exact report file templates may evolve; the system must support additional fields in future without redesign. This flexibility is explicitly expected in the scope.
 - 5. Oracle integration is not part of the initial core implementation, but the system must remain integration-ready.
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6. Detailed Functional Requirements

6.1 Authentication and Access Control

6.1.1 Login Screen

Purpose

Allow authorized users to securely access the platform.

Actors

Admin, Client User, OpCo User, Partner User

Fields

Field	Type	Required	Validation	Notes
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Email	Email/Text	Yes	Must be valid email format	Case-insensitive
Password	Password	Yes	Cannot be blank	Hidden input
Remember Me	Checkbox	No	Optional	Session behavior only

Actions

- Login
- Forgot Password
- Show/Hide Password

Validations

- Email is required
- Password is required
- Email format must be valid
- User account must exist
- User status must be Active
- Locked users shall not be allowed to log in

Error Messages

- Invalid email format
- Email is required
- Password is required
- Incorrect email or password
- Your account is inactive
- Your account is locked. Contact administrator

Business Rules

- Authentication shall be role-based
- After successful login, user shall be redirected to role-specific dashboard
- The system shall record last successful login timestamp
- Multiple failed login attempts shall trigger account lock based on configurable threshold

6.1.2 Forgot Password

Fields

Field	Type	Required	Validation
Email	Email/Text	Yes	Must exist in system

Business Rules

- Reset link or OTP shall be sent to registered email
- Reset token shall expire after configurable duration
- Expired or used token shall not be reusable

6.1.3 Change Password

Fields

Field	Type	Required	Validation
Current Password	Password	Yes	Must match existing password
New Password	Password	Yes	Min 8 chars, configurable complexity
Confirm Password	Password	Yes	Must match new password

Business Rules

- User shall not reuse last configurable number of passwords
- Password change shall be logged in audit trail

6.2 Dashboard Module

6.2.1 General Dashboard Requirements

Each role shall see a dashboard with information relevant to its access level.

Admin Dashboard Widgets

- total users
- active users
- inactive users
- pending submissions
- pending invoice reviews
- pending reconciliations
- overdue actions
- recent activity log summary

Client Dashboard Widgets

- pending OpCo reports
- pending partner reports
- pending reconciliations
- mismatches awaiting action
- invoices awaiting review
- invoices on hold
- collections pending from OpCos
- partner payments ready to process
- summary by month, partner, OpCo, and service
- alerts and reminders

OpCo Dashboard Widgets

- required monthly submission status
- uploaded reports
- invoices issued to OpCo
- payment status
- pending requests / reminders

Partner Dashboard Widgets

- required monthly report status
- invoice submission eligibility
- uploaded invoices
- payment status
- pending clarification items

Business Rules

- Dashboard data shall be filtered by user role and assigned entity
- Dashboard widgets shall be clickable where navigation is relevant
- Metrics shall reflect only authorized data
- Counts shall update in near real-time after transactions

6.3 Master Data and Configuration Module

6.3.1 OpCo Master

Fields:

- OpCo Code
- OpCo Name
- Country
- Currency
- Status
- Primary Contact Name
- Primary Contact Email
- Primary Contact Phone
- Remarks

Validations:

- OpCo Code unique
- OpCo Name required
- Email format valid if entered

6.3.2 Partner Master

Fields:

- Partner Code
- Partner Name
- Associated Services
- Default Currency
- Status
- Contact Name
- Contact Email
- Contact Phone
- Tax Registration fields if required later
- Remarks

Validations:

- Partner Code unique

- Partner Name required

6.3.3 Service Master

Fields:

- Service Code
- Service Name
- Status
- Description

Initial services include three revenue streams mentioned in scope.

6.3.4 Reminder Configuration

Fields:

- Reminder Type
 - Target Role
 - Trigger Event
 - Days Before Due Date
 - Days After Due Date
 - Repetition Frequency
 - Email Template
 - Is Active
-

6.4 User Management Module

6.4.1 Create/Edit User

Fields

Field	Type	Required	Validation
Full Name	Text	Yes	Max 150 chars
Email	Email	Yes	Unique, valid format
Mobile	Text	No	Max 20 chars

Role	Dropdown	Yes	Must be Admin, Client, OpCo, Partner
Assigned OpCo	Dropdown	Conditional	Required for OpCo user
Assigned Partner	Dropdown	Conditional	Required for Partner user
Status	Dropdown	Yes	Active, Inactive, Locked
Temporary Password	Password	Conditional	Required on create
Remarks	Textarea	No	Max configurable length

Business Rules

- Only Admin can create, edit, lock, unlock, deactivate, or delete users
- Email must be unique across all users
- OpCo users must be linked to one or more authorized OpCos
- Partner users must be linked to one or more authorized partners
- Client users may have broader access according to permission matrix

Delete Rules

- Hard delete should be restricted
 - Recommended behavior: soft delete or deactivate user
 - Historical audit data must remain intact even if user is inactive
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6.5 Report Management Module

The scope requires upload and view reports for OpCos and partners, month-wise, partner-wise, and service-wise consolidation and reconciliation.

6.5.1 Report Types

- OpCo Monthly Report
- Partner Monthly Report
- Client Consolidated Report
- Final RS Confirmation Report

6.5.2 Upload Report Screen

Actors

OpCo User, Partner User, Client User where permitted

Fields

Field	Type	Required	Validation	Notes
Report Type	Dropdown	Yes	Must be valid report type	Determines workflow
Reporting Month	Month	Yes	Cannot be future month	
Reporting Year	Year	Yes	Cannot be future year beyond current month context	
OpCo	Dropdown	Conditional	Required for OpCo report and related cases	Role-bound
Partner	Dropdown	Conditional	Required for Partner report and related cases	Role-bound
Service	Dropdown	Yes	Must exist in service master	
Report Reference No	Text	No	Unique if provided	Optional external reference
File Upload	File	Yes	Allowed types only	XLS, XLSX, CSV, PDF if needed
Remarks	Textarea	No	Max length	
Submission Date	System	Auto	Read only	
Submitted By	System	Auto	Read only	

Allowed File Types

- XLS
- XLSX
- CSV
- PDF only if business later approves document-only uploads

File Validations

- File type must be supported
- File size must not exceed configured maximum
- File name should not contain restricted characters if storage policy requires
- Virus/malware scan should pass where infrastructure supports it
- File must not be corrupted

Business Validations

- Reporting Month and Year are mandatory
- Report cannot be uploaded for a future reporting period
- Duplicate report for same Report Type + Reporting Month + Reporting Year + OpCo + Partner where applicable + Service shall not be allowed unless revision flow is enabled
- If revision flow is enabled, the previous report must remain retained with version history
- OpCo field is mandatory for OpCo report
- Partner field is mandatory for Partner report
- Service is mandatory
- Role-based restriction applies:
 - OpCo user can upload only for assigned OpCo
 - Partner user can upload only for assigned Partner
 - Client/Admin can upload based on permissions

Actions

- Upload
- Save Draft if enabled
- Cancel
- Download Template if available
- Preview Metadata

Post-Submission Behavior

- System stores file and metadata
- Status becomes Submitted
- Audit log entry created
- Notification triggered to Client users
- Submission timestamp stored

Report Statuses

- Draft
- Submitted
- Under Review

- Accepted
- Rejected
- Superseded if revised version uploaded
- Archived

Status Transition Rules

From	To	Allowed By	Condition
Draft	Submitted	Submitter	Required fields complete
Submitted	Under Review	Client/Admin	Review initiated
Under Review	Accepted	Client/Admin	Validation passed
Under Review	Rejected	Client/Admin	Validation or business issue
Accepted	Superseded	Client/Admin/System	Revised version approved
Rejected	Draft	Original submitter if allowed	Rework process enabled

6.5.3 View Reports Screen

Filters

- Report Type
- Reporting Month
- Reporting Year
- OpCo
- Partner
- Service
- Status
- Submitted By
- Date Range
- File Name / Reference Number

Grid Columns

- Report ID
- Report Type
- Reporting Month-Year
- OpCo
- Partner

- Service
- Version
- Status
- Submitted By
- Submission Date
- Last Updated
- Actions

Actions

- View metadata
- Download file
- Review
- Accept
- Reject
- Add remarks
- View history
- Compare versions where applicable

Business Rules

- Users shall only see authorized data
 - Accepted reports are eligible for reconciliation
 - Rejected reports require corrective re-submission where process is enabled
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6.6 Invoice Management Module

The scope requires invoice generation to OpCos, recording invoices received from partners, and tracking invoice and payment statuses by partner and OpCo. It also explicitly requires that partner invoices should be processed only after collections are received from respective OpCos.

6.6.1 Invoice Types

- OpCo Invoice generated by Client
 - Partner Invoice uploaded by Partner or recorded by Client
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6.6.2 OpCo Invoice Generation

Actors

Client User, Admin

Fields

Field	Type	Required	Validation
Reporting Month	Month	Yes	Cannot be future
Reporting Year	Year	Yes	Valid year
OpCo	Dropdown	Yes	Must be active OpCo
Service	Dropdown	Yes	Must exist
Linked Reconciliation ID	Lookup	Yes	Must be finalized or approved for invoicing
Invoice Number	Text	Yes	Unique
Invoice Date	Date	Yes	Cannot be future
Currency	Dropdown	Yes	Valid currency
Invoice Amount	Decimal	Yes	Must be > 0
Due Date	Date	No/Configurable	Must be >= Invoice Date
Attachment / PDF	File/System	Optional initially	Generated or uploaded
Remarks	Textarea	No	Max length

Business Rules

- OpCo invoice shall be generated from consolidated or confirmed values
- Invoice shall not be generated if reconciliation is incomplete unless authorized override is allowed
- Invoice number must be unique
- Invoice amount must be based on approved calculation source
- Generated invoice may be system-generated PDF in later phase

Statuses

- Draft
- Generated

- Sent to OpCo
- Acknowledged
- Partially Paid
- Fully Paid
- Overdue
- Cancelled

6.6.3 Partner Invoice Upload / Record

Actors

Partner User, Client User, Admin

Fields

Field	Type	Required	Validation
Partner	Dropdown	Yes	Must be valid partner
Related OpCo	Dropdown	Yes	Must be valid and linked contextually
Reporting Month	Month	Yes	Cannot be future
Reporting Year	Year	Yes	Valid year
Service	Dropdown	Yes	Must exist
Invoice Number	Text	Yes	Unique per partner within configurable scope
Invoice Date	Date	Yes	Cannot be future
Currency	Dropdown	Yes	Valid currency
Invoice Amount	Decimal	Yes	Greater than zero
File Upload	File	Yes	PDF preferred, other configurable
Remarks	Textarea	No	Max length
Submitted By	System	Auto	Read only
Submission Timestamp	System	Auto	Read only

Validations

- Invoice number required
- Invoice date required
- Invoice amount greater than zero
- Duplicate invoice number for same partner and same reporting period not allowed unless revision type supported
- File required if partner is uploading invoice
- Currency required
- Related reporting period required

Critical Business Rule

A partner invoice shall be eligible for processing only after collections from the respective OpCo are received. This rule is explicitly stated in the scope and is one of the most important controls in the system.

Invoice Processing Statuses

- Uploaded
- Under Review
- On Hold Awaiting OpCo Collection
- Eligible for Processing
- Approved for Payment
- Partially Paid
- Paid
- Rejected
- Cancelled

Status Transition Rules

From	To	Allowed By	Condition
Uploaded	Under Review	Client/Admin	Initial review started
Under Review	On Hold Awaiting OpCo Collection	Client/Admin/System	Collection not received
Under Review	Eligible for Processing	Client/Admin/System	Collection confirmed
Eligible for Processing	Approved for Payment	Client/Admin	Review complete
Approved for Payment	Partially Paid	Client/Admin	Partial settlement done

Approved for Payment	Paid	Client/Admin	Full settlement done
Under Review	Rejected	Client/Admin	Validation/business issue
Any non-final	Cancelled	Admin/Authorized Client	Valid cancellation reason

6.6.4 View Invoices Screen

Filters

- Invoice Type
- Invoice Number
- OpCo
- Partner
- Reporting Month-Year
- Service
- Status
- Payment Status
- Date Range
- Currency

Columns

- Invoice ID
- Invoice Type
- Invoice Number
- Party Name
- Related OpCo
- Related Partner
- Service
- Reporting Period
- Amount
- Currency
- Status
- Payment Status
- Invoice Date
- Due Date
- Last Updated

Actions

- View
 - Download
 - Edit where permitted
 - Update status
 - Add remarks
 - Link collection / payment
 - View history
 - Export
-

6.7 Reconciliation Management Module

This is the core operational module. The scope requires reconciliation of OpCo reports with partner reports month-wise, partner-wise, and service-wise, mismatch alignment, communication, and final confirmation of numbers.

6.7.1 Reconciliation Inputs

Reconciliation shall be driven at minimum by:

- Reporting Month
- Reporting Year
- OpCo
- Partner where applicable
- Service

6.7.2 Reconciliation Workbench Screen

Filters / Inputs

Field	Type	Required	Validation
Reporting Month	Month	Yes	Must exist in submitted data
Reporting Year	Year	Yes	Must exist in submitted data
OpCo	Dropdown	Optional / Often required	Filter based on role
Partner	Dropdown	Optional / Often required	Filter based on role

Service	Dropdown	Optional / Often required	Based on business selection
Reconciliation Status	Dropdown	No	Valid status
Show Exceptions Only	Checkbox	No	Optional

Actions

- Load Reports
- Run Reconciliation
- View Matched Records
- View Mismatches
- Add Clarification Note
- Record Adjustment
- Save Progress
- Finalize
- Reopen if permitted
- Export Results
- Send Notification

6.7.3 Reconciliation Data Elements

At minimum the reconciliation shall compare:

- quantity or transaction counts where relevant
- revenue amount
- service-level metrics as defined by report template
- any future configured additional fields

6.7.4 Matching Logic

Matching shall occur using the reconciliation key. Initial base key:

- Reporting Month-Year
- OpCo
- Partner
- Service

If line-level templates are introduced later, additional keys may include:

- transaction identifier
- campaign identifier

- API batch identifier
- region
- subservice
- custom business field

6.7.5 Reconciliation Outcomes

For each comparable record or aggregate:

- Matched
- Mismatch
- Missing on OpCo side
- Missing on Partner side
- Pending Review
- Adjusted
- Confirmed

6.7.6 Reconciliation Statuses

- Not Started
- In Progress
- Mismatch Found
- Pending Clarification
- Ready for Finalization
- Finalized
- Reopened
- Cancelled

6.7.7 Business Rules

1. Reconciliation cannot begin unless required source reports exist.
2. Reconciliation shall compare data month-wise, partner-wise, and service-wise.
3. Missing reports shall block normal reconciliation processing and trigger reminder/exception workflow.
4. Mismatches shall be visible distinctly from matched records.
5. Users shall be able to record remarks or clarifications against mismatches.
6. Final numbers shall be confirmable after mismatch resolution or authorized override.
7. Finalized reconciliations shall become read-only except for authorized reopen action.
8. Reopened reconciliations must retain full change history.
9. Final confirmation may result in generation of RS report / final confirmation output.
10. Finalized reconciliation values shall become source for downstream invoicing where applicable.

6.7.8 Validations

- Reporting period must be selected
- At least one valid report set must exist
- Reconciliation cannot finalize if mandatory exceptions remain unresolved unless override privilege exists
- Adjustment amount, if entered, must be numeric and justified with remarks
- Finalization reason or override reason must be mandatory where mismatch remains unresolved
- Only authorized Client/Admin users may finalize or reopen

6.7.9 Reconciliation Detail Grid

Columns may include:

- Line / Record ID
- OpCo
- Partner
- Service
- Reporting Period
- OpCo Value
- Partner Value
- Difference
- Difference %
- Status
- Adjustment
- Final Confirmed Value
- Remarks
- Last Action By
- Last Action Date

6.7.10 Finalization Rules

- All required source reports accepted or approved for use
 - Review completed
 - Any mismatch either resolved, adjusted, or approved under override
 - Final values stored
 - Finalization timestamp and user recorded
 - Finalization audit preserved
-

6.8 Collection and Payment Tracking Module

The scope requires following up collections from OpCos and preparing OpCo-wise / Partner-wise reports for collections received and payments to be made.

6.8.1 Collection Tracking

Fields

Field	Type	Required	Validation
OpCo Invoice	Lookup	Yes	Must exist
OpCo	Auto/Lookup	Yes	Derived
Amount Invoiced	Decimal	Auto	Read only
Amount Received	Decimal	Yes	>= 0
Collection Date	Date	Yes	Cannot be future
Collection Reference	Text	No	Optional unique reference
Currency	Auto/Override by rules	Yes	Must match or support conversion policy
Collection Status	System/Dropdown	Yes	Derived or controlled
Remarks	Textarea	No	Max length

Collection Statuses

- Not Received
- Partially Received
- Fully Received
- Reconciled
- Disputed

Business Rules

- Full collection means amount received equals invoice amount
 - Partial collection means amount received less than invoice amount but greater than zero
 - Partner invoice eligibility depends on relevant collection received rule; exact threshold may be full collection only unless business config allows proportional release
 - Any manual adjustment in collection status must be logged
-

6.8.2 Partner Payment Tracking

Fields

Field	Type	Required	Validation
Partner Invoice	Lookup	Yes	Must exist
Amount Approved	Decimal	Yes	> 0 and <= outstanding eligible amount
Amount Paid	Decimal	Yes	>= 0
Payment Date	Date	Conditional	Required when amount paid > 0
Payment Reference	Text	No	Optional
Payment Status	System/Dropdown	Yes	Derived or controlled
Remarks	Textarea	No	Max length

Payment Statuses

- Not Ready
- Ready for Payment
- Partially Paid
- Fully Paid
- On Hold
- Cancelled

Business Rules

- Partner payment cannot be approved unless invoice is eligible
- Eligibility is controlled by OpCo collection status rule
- Partial partner payments must be supported

- Outstanding balances must remain visible
-

6.9 Notification and Reminder Module

The scope explicitly includes sending email notifications to OpCos and partners and setting reminder durations.

6.9.1 Notification Types

- report submission reminder
- report uploaded confirmation
- invoice uploaded confirmation
- report rejected notification
- mismatch clarification request
- reconciliation finalized notification
- OpCo invoice sent notification
- partner invoice on hold notification
- partner invoice eligible notification
- overdue collection reminder
- overdue payment reminder
- user creation / password reset notification

6.9.2 Reminder Settings Fields

Field	Type	Required	Validation
Reminder Name	Text	Yes	Unique
Target Role	Dropdown	Yes	Valid role
Trigger Event	Dropdown	Yes	Must be supported event
Trigger Timing	Dropdown	Yes	Before due date / after due date / instant
Days Offset	Integer	Yes	>= 0
Repeat Every X Days	Integer	No	>= 1 if repeat enabled
Email Template	Dropdown	Yes	Must exist

Active

Checkbox Yes

Boolean

6.9.3 Business Rules

- Only Admin or authorized Client users can configure reminder settings
 - Reminders shall not be sent for completed actions
 - Duplicate reminders within same configured window should be prevented
 - Notification logs shall record recipient, event, template, date-time, and delivery status
-

6.10 Search, Sort, Filter, and Export Module

The scope requires search capability with export to Excel and support for search, sort, and filter across the application.

6.10.1 Common Search Features

The system shall provide search, sort, filter, and export for:

- reports
- invoices
- reconciliation records
- collections
- partner payments
- users
- audit logs where permitted

6.10.2 Common Filters

- reporting month
- reporting year
- OpCo
- partner
- service
- status
- payment status
- date ranges

- amount ranges
- invoice number
- report type
- user / submitter

6.10.3 Export Rules

- Export format initially Excel, with CSV optional
 - Export shall respect applied filters
 - Export shall only include authorized records
 - Export action shall be auditable for sensitive datasets if required
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6.11 Administration Module

The scope requires create/edit/delete users, set email notification settings, and reminder settings.

6.11.1 Administration Features

- manage users
 - manage OpCo master
 - manage partner master
 - manage service master
 - manage email templates
 - manage reminder settings
 - manage status lookup values where configurable
 - view audit logs
 - configure report template behavior if future phase supports it
-

6.12 Audit and Activity Log Module

6.12.1 Audit Requirements

The system shall log at minimum:

- login success and failure

- password change
- user creation, edit, status change
- report upload, edit, review, accept, reject
- invoice upload, generation, review, status update
- reconciliation start, save, adjustment, finalize, reopen
- collection updates
- partner payment updates
- reminder and notification configuration changes
- export actions for sensitive datasets if enabled

6.12.2 Audit Data Fields

- Audit ID
- Module Name
- Action Type
- Entity ID
- Entity Type
- Old Value snapshot where applicable
- New Value snapshot where applicable
- Action By User ID
- Action By User Name
- Action Date-Time
- IP / client metadata if infrastructure supports
- Remarks / reason where required

6.12.3 Business Rules

- Audit records must not be editable by normal users
 - Audit data retention period shall be configurable
 - Finalized transactional records must preserve complete history
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7. Screen-Level Specifications

7.1 Login Screen

Already defined in section 6.1.1.

7.2 Dashboard Screen

Each role lands on dashboard after login.

Buttons may include:

- View Pending Reports
- View Pending Invoices
- Run Reconciliation
- View Collections
- View Payments
- Open Notifications
- Export Summary

7.3 Upload Report Screen

Already defined in section 6.5.2.

7.4 View Reports Screen

Already defined in section 6.5.3.

7.5 Upload Invoice Screen

Screen shall contain partner invoice fields defined in 6.6.3.

Buttons:

- Submit
- Save Draft if enabled
- Cancel
- Download Attachment
- View Related Reconciliation if permitted

7.6 View Invoices Screen

Already defined in 6.6.4.

7.7 Reconciliation Workbench Screen

Contains:

- filter panel
- summary card
- matched/mismatch tabs
- detail grid
- remarks / clarification panel
- actions for save, finalize, export, notify

7.8 Collection Tracking Screen

Contains:

- invoice lookup
- collection amount recording
- status auto-evaluation
- history panel

7.9 Partner Payment Screen

Contains:

- eligible invoice list
- approval / payment form
- outstanding balance summary
- payment history

7.10 User Management Screen

Contains:

- search/filter
- create/edit/deactivate/lock user actions
- role and entity assignment

7.11 Reminder Settings Screen

Contains:

- reminder list
- create/edit reminder
- assign template
- activate/deactivate

7.12 Audit Log Screen

Contains:

- date filter
 - module filter
 - action filter
 - user filter
 - entity search
 - detail viewer
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8. Detailed Business Rules

8.1 Report Rules

1. Reports are submitted monthly.
2. Reports must be tagged by reporting period, service, and entity context.
3. Duplicate report entries for the same business key are not allowed unless revision workflow is enabled.
4. Future period submissions are not allowed.
5. Only authorized users may upload reports for their entity.
6. Accepted or approved reports are used for reconciliation.
7. Rejected reports require correction and resubmission.

8.2 Reconciliation Rules

1. Reconciliation compares OpCo reports with partner reports.
2. Reconciliation shall support month-wise, partner-wise, and service-wise processing.
3. Missing data must be flagged as exception.
4. Mismatches must be visible and actionable.
5. Final confirmation is required before downstream settlement actions.

6. Finalized reconciliations are locked unless reopened by authorized users.
7. Reopening requires reason capture.

8.3 Invoice Rules

1. OpCo invoices are based on confirmed consolidated values.
2. Partner invoices may be uploaded or recorded, but shall only be processed after relevant OpCo collections are received.
3. Duplicate invoice number by same partner in same reporting context is not allowed.
4. Cancelled invoices shall not be considered for payment.
5. Partial payment must be supported where business allows.

8.4 Collection and Payment Rules

1. Collection from OpCo must be tracked against invoice.
2. Partner payment eligibility depends on collection rule.
3. Outstanding balances must remain visible.
4. Payment actions require authorized roles.

8.5 Notification Rules

1. Reminders shall be configurable.
2. Reminders shall be sent only for pending actions.
3. Event-driven notifications shall be logged.
4. Duplicate reminder flooding shall be prevented.

8.6 Security and Access Rules

1. Users can access only authorized entities and functions.
 2. UI visibility does not replace backend authorization.
 3. Sensitive actions must be audit logged.
 4. Locked or inactive users cannot access the system.
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9. Validations

9.1 Common Validations

- required field validation
- email format validation
- number and decimal validation
- date range validation
- future date prevention where relevant
- uniqueness validation
- role-based field validation
- file type and size validation

9.2 Conditional Validations

- Assigned OpCo required if role is OpCo User
- Assigned Partner required if role is Partner User
- OpCo required on OpCo report
- Partner required on Partner report
- Related collection confirmation required before partner invoice becomes eligible
- Payment date required if payment amount entered
- Reopen reason required when finalized reconciliation is reopened
- Rejection reason required when report or invoice is rejected

9.3 Error Handling Validations

- invalid file format
 - corrupted upload file
 - missing template-required columns if structured parsing is enabled
 - duplicate records
 - unauthorized action
 - stale status transition
 - concurrent update conflict if two users edit same record
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10. Status Transition Tables

10.1 Report Status

From	To	Allowed By	Condition
Draft	Submitted	Submitter	Mandatory fields complete

Submitted	Under Review	Client/Admin	Review started
Under Review	Accepted	Client/Admin	Valid and approved
Under Review	Rejected	Client/Admin	Issues found
Accepted	Superseded	Client/Admin	Revised version accepted
Rejected	Draft	Submitter	Resubmission allowed

10.2 Reconciliation Status

From	To	Allowed By	Condition
Not Started	In Progress	Client/Admin	Source data loaded
In Progress	Mismatch Found	System	Differences detected
In Progress	Ready for Finalization	System/User	No blocking issues
Mismatch Found	Pending Clarification	Client/Admin	Clarification requested
Pending Clarification	Ready for Finalization	Client/Admin	Resolved/adjusted
Ready for Finalization	Finalized	Client/Admin	Authorized finalization
Finalized	Reopened	Admin/Authorized Client	Reason mandatory

10.3 OpCo Invoice Status

From	To	Allowed By	Condition
Draft	Generated	Client/Admin	Values available
Generated	Sent to OpCo	Client/Admin/System	Issue action completed
Sent to OpCo	Acknowledged	Client/Admin	Acknowledgement recorded

Sent to OpCo	Partially Paid	Client/Admin	Partial collection received
Sent to OpCo	Fully Paid	Client/Admin	Full collection received
Partially Paid	Fully Paid	Client/Admin	Balance cleared
Any active	Cancelled	Admin/Authorized Client	Reason provided

10.4 Partner Invoice Status

From	To	Allowed By	Condition
Uploaded	Under Review	Client/Admin	Review started
Under Review	On Hold Awaiting OpCo Collection	Client/Admin/System	Collection not received
Under Review	Eligible for Processing	Client/Admin/System	Collection received
Eligible for Processing	Approved for Payment	Client/Admin	Approved
Approved for Payment	Partially Paid	Client/Admin	Partial settlement
Approved for Payment	Paid	Client/Admin	Full settlement
Under Review	Rejected	Client/Admin	Invalid or disputed
Any non-final	Cancelled	Admin/Authorized Client	Reason provided

11. Process Flows

11.1 Monthly Report Submission Process

Primary Flow

1. System calculates due reports for the reporting month.
2. System sends reminder emails to OpCos and partners based on reminder settings.
3. User logs in and opens Upload Report screen.
4. User selects report type, reporting month-year, relevant entity fields, service, and file.
5. System validates metadata and file.
6. On success, system stores file and report metadata.
7. System sets report status to Submitted.
8. System records audit log.
9. System sends confirmation or alert notification to Client team.
10. Client reviews submitted report.

Alternate Flow A: Duplicate Submission

1. User uploads a report with same business key.
2. System detects duplicate.
3. If revision workflow disabled, upload is blocked.
4. If revision workflow enabled, user may upload as revised version based on permission.

Alternate Flow B: Invalid File

1. User uploads unsupported or corrupted file.
 2. System rejects upload.
 3. Error message displayed.
 4. No submission record created, or temporary failed attempt logged.
-

11.2 Report Review Process

1. Client opens submitted report.
 2. Client checks metadata and attachment.
 3. Client either accepts or rejects.
 4. If rejected, rejection reason mandatory.
 5. Notification may be sent back to submitter.
 6. If accepted, report becomes available for reconciliation.
-

11.3 Reconciliation Process

Primary Flow

1. Client selects reporting period and relevant filters.
2. System retrieves accepted source reports.
3. System compares OpCo and partner data based on reconciliation key.
4. Matched records are marked Matched.
5. Differences are marked Mismatch.
6. Missing side data is marked Exception.
7. Client reviews mismatches.
8. Client records remarks, clarification notes, or adjustments.
9. Once all required items are resolved, system allows finalization.
10. Client finalizes reconciliation.
11. Final values are stored.
12. Audit log is created.
13. Final confirmation output becomes available for invoicing / partner communication.

Alternate Flow A: Missing Report

1. One side report is missing.
2. System blocks full reconciliation.
3. Exception status shown.
4. Reminder or notification can be triggered.

Alternate Flow B: Unresolved Mismatch at Finalization

1. Client attempts finalize.
 2. System checks unresolved blocking mismatches.
 3. Finalization blocked unless override privilege exists.
 4. If override allowed, reason mandatory and audit logged.
-

11.4 OpCo Invoice Generation Process

1. Client opens finalized reconciliation dataset.
2. Client selects records for invoice generation.
3. System calculates invoice amount from confirmed values.
4. User enters or confirms invoice metadata.

5. System validates uniqueness and amount.
 6. Invoice generated and stored.
 7. Invoice status becomes Generated.
 8. Client sends invoice to OpCo.
 9. Status updated to Sent to OpCo.
-

11.5 Collection Tracking Process

1. Client receives payment or collection advice from OpCo.
 2. Client opens collection tracking.
 3. Client selects related OpCo invoice.
 4. Client enters amount received and date.
 5. System recalculates collection status.
 6. If full amount received, system marks Fully Paid.
 7. Partner invoice eligibility is recalculated for linked context.
-

11.6 Partner Invoice Processing Process

1. Partner uploads invoice or Client records it.
 2. System validates invoice details.
 3. System checks linked OpCo collection rule.
 4. If collection not received, status becomes On Hold Awaiting OpCo Collection.
 5. If collection received, status becomes Eligible for Processing.
 6. Client reviews invoice.
 7. Client approves for payment.
 8. Payment details recorded as payments occur.
 9. Status changes to Partially Paid or Paid.
-

11.7 Reminder Process

1. Scheduler identifies pending due actions.
2. System checks reminder configuration.
3. System ensures action not already completed.
4. System ensures duplicate reminder threshold not violated.

5. Email notification sent.
 6. Notification log entry recorded.
-

12. Role Permission Matrix

Function	Admin	Client User	OpCo User	Partner User
Login	Yes	Yes	Yes	Yes
Change Password	Yes	Yes	Yes	Yes
View Dashboard	Yes	Yes	Yes	Yes
Upload OpCo Report	Optional	Optional	Yes	No
Upload Partner Report	Optional	Optional	No	Yes
View Own Reports	Yes	Yes	Yes	Yes
View All Reports	Yes	Yes	No	No
Review Reports	Yes	Yes	No	No
Accept/Reject Reports	Yes	Yes	No	No
Generate OpCo Invoice	Yes	Yes	No	No
Upload Partner Invoice	Optional	Optional	No	Yes
Review Partner Invoice	Yes	Yes	No	No
Update Invoice Status	Yes	Yes	No	No
Track Collections	Yes	Yes	No	No
Approve Partner Payment	Yes	Yes	No	No
Perform Reconciliation	Yes	Yes	No	No
Finalize Reconciliation	Yes	Yes	No	No
Reopen Reconciliation	Yes	Restricted	No	No
Configure Reminders	Yes	Optional	No	No

Manage Users	Yes	No	No	No
View Audit Logs	Yes	Restricted	No	No
Export Search Results	Yes	Yes	Limited own data	Limited own data

13. Reporting Requirements

13.1 Standard Reports

- OpCo-wise collection report
- Partner-wise payment report
- Month-wise reconciliation summary
- Service-wise reconciliation summary
- Pending report submission report
- Pending invoice review report
- Invoices on hold report
- Mismatch report
- Aging of overdue collections
- Partner payment readiness report
- Dashboard KPI reports

13.2 Adhoc Reporting

The system shall support customized reports across available data fields, as expected in the scope.

13.3 Export

- Excel export mandatory
 - CSV optional
 - Filters applied before export must be preserved
-

14. Non-Functional Requirements

14.1 Performance

- Normal dashboard and list loads should complete within acceptable business response times
- Search and export performance must remain reasonable for expected data volumes
- File uploads should provide progress or completion feedback

14.2 Scalability

- System should support current seven OpCos and more than 35 partners, with future growth capacity.
- Architecture should support additional fields and new functionalities without major redesign.

14.3 Security

- Role-based authorization
- Secure password storage
- Protected file access
- Audit logging of critical actions
- Session control and timeout

14.4 Usability

- Responsive interface required by scope.
- Clear menus and navigation
- Simple, business-friendly forms and workflows

14.5 Reliability

- Data must not be lost during normal operations
- File and transaction save operations must confirm success or failure
- Critical status changes must be consistent and traceable

14.6 Maintainability

- Configurable masters and reminder settings
 - Extendable field definitions
 - Prepared structure for ERP integration
-

15. Integration Requirements

15.1 Email Integration

The system shall integrate with an email service for notifications and reminders.

15.2 ERP Integration Readiness

The system should be able to connect with Oracle ERP at a later stage.

Integration readiness requirements:

- stable unique transaction identifiers
 - exportable / mappable financial records
 - clean master data relationships
 - status-based downstream readiness
-

16. Exception Handling

16.1 Upload Exceptions

- unsupported file type
- corrupted file
- duplicate report/invoice
- missing required metadata
- unauthorized upload

16.2 Reconciliation Exceptions

- missing side report
- unparseable structured data if parsing enabled
- unresolved mismatch
- reopened finalized record conflict

16.3 Payment Exceptions

- partner invoice uploaded before collection eligibility

- payment exceeds approved amount
 - currency mismatch scenario if later enabled
 - duplicate payment reference if uniqueness enforced
-

17. Open Configuration Items for Business Confirmation

These are not gaps in the SRS, but configurable decisions the business may finalize during implementation:

1. maximum file size limit
 2. exact supported file formats by module
 3. password policy strictness
 4. lockout threshold
 5. whether draft mode is enabled for reports/invoices
 6. whether partial OpCo collection can proportionally release partner payment eligibility
 7. exact reminder frequency and escalation model
 8. whether system-generated invoice PDFs are in phase 1 or later phase
 9. whether line-level parsing is required initially or only metadata/file management in phase 1
 10. audit retention duration
-

18. Conclusion

This SRS defines the Reconciliation & Reporting Tool at a detailed enterprise level, including:

- screen-level behavior
- field-level specifications
- validations
- status lifecycles
- business rules
- end-to-end process flows
- reporting and audit requirements
- access control and future integration readiness